Status	Finished
Started	Thursday, 29 October 2026, 5:55 PM
Completed	Thursday, 29 October 2026, 6:02 PM
Duration	6 mins 54 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. There must be a mechanism to access middle element directly

b. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

c. Must use a sorted array

d. Binary search algorithm is not efficient when the data elements more than 1500 ✓

Question **2**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(1)

b. O(log n) ✓

c. O🥐

d. O(n log n)

Question **3**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

a. Compare the target element with the first element in the list

b. Divide the list into two equal parts

c. Sort the list

d. Compare the target element with the middle element in the list ✓

Question **4**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. In the middle of the list
b. In the left sublist
c. In the right sublist ✓
d. At the beginning of the list

Question **5**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search can find multiple instances of the target element
b. Binary search works on unsorted lists
c. Binary search has a lower time complexity on large, sorted lists ✓
d. Binary search does not require dividing the list

Question **6**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search divides the list into halves.
b. Linear search requires the list to be sorted.
c. Linear search is more efficient than binary search on large lists.
d. Linear search can be applied to both sorted and unsorted lists. ✓

Question **7**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It requires the list to be sorted ✓
b. It is slower than linear search for small lists
c. It does not work with negative numbers
d. It can only be applied to large lists

Question **8**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. The last element is returned
b. An error is raised
c. The search is considered unsuccessful ✓
d. The first element is returned

Question **9**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

a. Linear search ✓
b. Binary search
c. Jump search
d. Hashing

Question **10**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Mining
b. Searching ✓
c. Discovering
d. Finding

Question **11**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

- a. Binary ✓
- b. Both (1) & (3)
- c. Hash
- d. Linear

Question **12**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

- a. The search continues from the beginning
- b. The search continues in the left or right sublist ✓
- c. The list is sorted
- d. The search stops

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

- a. Merge search
- b. Linear search ✓
- c. Bubble search
- d. Quick search

Question **14**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

- a. 1
- b. n ✓
- c. n/2
- d. log n

Question **15**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By sorting the list first

b. By comparing each element in the list sequentially ✓

c. By using a hash function

d. By dividing the list into halves